

Problem 35: Application to Differential Equations

Example 103: Solve the following Initial Value Problem (IVP) by Laplace Transform

$$Y'' + Y = t, \quad Y(0) = 1, \quad Y'(0) = -2$$

Solution

$$\text{Let } Y = f(t)$$

$$Y' = f'(t)$$

$$Y'' = f''(t)$$

Given,

$$Y'' + Y = t$$

$$\text{That is } \frac{d^2 Y}{dt^2} + Y = t$$

Taking the Laplace transform of both sides of the differential equation and using the given conditions, we have

$$Y'' + Y = t$$

$$L\{Y''\} + L\{Y\} = L\{t\}$$

We have,

$$L(f''(t)) = s^2 L\{f(t)\} - s f(0) - f'(0)$$

$$L(Y'') = s^2 L\{Y\} - s f(0) - f'(0)$$

Therefore,

$$L\{Y''\} + L\{Y\} = L\{t\}$$

$$s^2 L\{Y\} - s f(0) - f'(0) + L\{Y\} = L\{t\}$$

$$s^2 y - s f(0) - f'(0) + y = L\{t\}$$

$$[\text{let, } L\{Y\} = y]$$

$$s^2 y - s f(0) - f'(0) + y = \frac{1}{s^2}$$

$$[\text{let, } L\{t\} = \frac{1}{s^2}]$$

$$s^2 y - s.1 - (-2) + y = \frac{1}{s^2}$$

$$[\text{Given, } Y(0) = f(0) = 1 \quad Y'(0) = f'(0) = -2]$$

$$s^2 y + y - s.1 + 2 = \frac{1}{s^2}$$

$$s^2 y + y - s.1 + 2 - \frac{1}{s^2} = 0$$

$$y(s^2 + 1) - s + 2 - \frac{1}{s^2} = 0$$

$$y(s^2 + 1) = s - 2 + \frac{1}{s^2}$$

$$y = \frac{s-2}{s^2+1} + \frac{1}{s^2(s^2+1)}$$

$$y = \frac{s}{s^2+1} - \frac{2}{s^2+1} + \frac{1}{s^2} - \frac{1}{(s^2+1)}$$

$$y = \frac{s}{s^2+1} + \frac{1}{s^2} - \frac{3}{s^2+1}$$

$$\therefore L\{Y\} = y = \frac{s}{s^2+1} + \frac{1}{s^2} - \frac{3}{s^2+1}$$

$$\therefore Y = L^{-1}(y) = L^{-1}\left[\frac{s}{s^2+1} + \frac{1}{s^2} - \frac{3}{s^2+1}\right]$$

$$\therefore Y = L^{-1}(y) = L^{-1}\left[\frac{s}{s^2+1}\right] + L^{-1}\left[\frac{1}{s^2}\right] - 3L^{-1}\left[\frac{1}{s^2+1}\right]$$

$$\therefore Y = L^{-1}(y) = \cos t + t - 3\sin t \quad \text{Answer}$$

Proof:

$$\therefore Y = \cos t + t - 3\sin t$$

$$\therefore Y' = -\sin t + 1 - 3\cos t$$

$$\therefore Y'' = -\cos t + 0 + 3\sin t$$

$$\therefore Y'' + Y = -\cos t + 0 + 3\sin t + \cos t + t - 3\sin t$$

$$\therefore Y'' + Y = t$$

Again,

$$\therefore Y = \cos t + t - 3\sin t$$

$$\therefore Y(0) = \cos 0 + 0 - 3\sin 0$$

$$\therefore Y(0) = 1$$

Again

$$\therefore Y' = -\sin t + 1 - 3\cos t$$

$$\therefore Y'(0) = -\sin 0 + 1 - 3\cos 0$$

$$\therefore Y'(0) = 0 + 1 - 3 \cdot 1$$

$$\therefore Y'(0) = -2$$

Example 104: Solve the following Initial Value Problem (IVP) by Laplace Transform

$$Y'' + 4Y = 12t \quad Y(0) = 0 \quad Y'(0) = 7$$

Solution

Given,

$$Y'' + 4Y = 12t$$

Taking the Laplace transform of both sides of the differential equation and using the given conditions, we have

$$Y'' + 4Y = 12t$$

$$L\{Y''\} + 4L\{Y\} = 12L\{t\}$$

We have,

$$L\{f''(t)\} = s^2 L\{f(t)\} - s f(0) - f'(0)$$

$$L\{Y''\} + 4L\{Y\} = 12L\{t\}$$

$$s^2 L\{Y\} - s f(0) - f'(0) + 4L\{Y\} = 12L\{t\}$$

$$s^2y - sf(0) - f'(0) + 4y = 12L\{t\}$$

$$[let, L\{Y\} = y]$$

$$s^2y - sf(0) - f'(0) + 4y = 12\frac{1}{s^2}$$

$$[let, L\{t\} = \frac{1}{s^2}]$$

$$s^2y - s.0 - 7 + 4y = 12\frac{1}{s^2}$$

$$[Given, Y(0) = 0 \quad Y'(0) = 7]$$

$$s^2y - 7 + 4y = \frac{12}{s^2}$$

$$s^2y + 4y - 7 - \frac{12}{s^2} = 0$$

$$y(s^2 + 4) - 7 - \frac{12}{s^2} = 0$$

$$y(s^2 + 4) = 7 + \frac{12}{s^2}$$

$$y = \frac{7}{(s^2 + 4)} + \frac{12}{s^2(s^2 + 4)}$$

$$y = \frac{7}{(s^2 + 4)} + 3 \times \frac{4}{s^2(s^2 + 4)}$$

$$y = \frac{7}{(s^2 + 4)} + 3 \times \left[\frac{1}{s^2} - \frac{1}{(s^2 + 4)} \right]$$

$$y = \frac{7}{(s^2 + 4)} + \left[\frac{3}{s^2} - \frac{3}{(s^2 + 4)} \right]$$

$$\therefore L\{Y\} = y = \frac{7}{(s^2 + 4)} + \left[\frac{3}{s^2} - \frac{3}{(s^2 + 4)} \right]$$

$$\therefore Y = L^{-1}(y) = L^{-1} \left[\frac{7}{(s^2 + 4)} + \left[\frac{3}{s^2} - \frac{3}{(s^2 + 4)} \right] \right]$$

$$\therefore Y = L^{-1}(y) = L^{-1} \left[\frac{7}{(s^2 + 4)} \right] + L^{-1} \left[\frac{3}{s^2} - \frac{3}{(s^2 + 4)} \right]$$

$$\therefore Y = L^{-1}(y) = 7L^{-1} \left[\frac{1}{(s^2 + 4)} \right] + 3L^{-1} \left[\frac{1}{s^2} - \frac{1}{(s^2 + 4)} \right]$$

$$\therefore Y = L^{-1}(y) = 7L^{-1} \left[\frac{1}{(s^2 + 4)} \right] + 3L^{-1} \left[\frac{1}{s^2} \right] - 3L^{-1} \left[\frac{1}{(s^2 + 4)} \right]$$

$$\therefore Y = L^{-1}(y) = 4L^{-1} \left[\frac{1}{(s^2 + 4)} \right] + 3L^{-1} \left[\frac{1}{s^2} \right]$$

$$\therefore Y = L^{-1}(y) = 4 \frac{1}{2} L^{-1} \left[\frac{2}{(s^2 + 4)} \right] + 3L^{-1} \left[\frac{1}{s^2} \right]$$

$$\therefore Y = L^{-1}(y) = 4 \frac{1}{2} \sin 2t + 3t$$

$$\therefore Y = L^{-1}(y) = 2 \sin 2t + 3t$$

Example 105: Solve the following Initial Value Problem (IVP) by Laplace Transform

$$Y'' - 3Y' + 2Y = 4e^{2t} \quad Y(0) = -3 \quad Y'(0) = 5$$

Solution

$$Y = f(t)$$

Given,

$$Y'' - 3Y' + 2Y = 4e^{2t}$$

Taking the Laplace transform of both sides of the differential equation and using the given conditions, we have

$$Y'' - 3Y' + 2Y = 4e^{2t}$$

$$L\{Y''\} - 3L\{Y'\} + 2L\{Y\} = 4L\{e^{2t}\}$$

We have,

$$L\{f''(t)\} = s^2 L\{f(t)\} - s f(0) - f'(0)$$

And

$$\therefore L\{f'(t)\} = sL\{f(t)\} - f(0)$$

$$\therefore L\{Y''\} - 3L\{Y'\} + 2L\{Y\} = 4L\{e^{2t}\}$$

$$s^2 L\{f(t)\} - s f(0) - f'(0) - 3[sL\{f(t)\} - f(0)] + 2y = 4 \frac{1}{s-2} \quad [\text{let, } L\{Y\} = y \text{ \& } L(e^{at}) = \frac{1}{s-a}]$$

$$s^2 L\{Y\} - s f(0) - f'(0) - 3[sL\{Y\} - f(0)] + 2y = 4 \frac{1}{s-2} \quad [Y = f(t)]$$

$$s^2 y - s f(0) - f'(0) - 3[sy - f(0)] + 2y = 4 \frac{1}{s-2} \quad [\text{let, } L\{Y\} = y]$$

$$s^2 y - s f(0) - f'(0) - 3sy + 3f(0) + 2y = 4 \frac{1}{s-2}$$

$$s^2 y - s\{-3\} - 5 - 3sy + 3(-3) + 2y = 4 \frac{1}{s-2}$$

$$s^2 y + 3s - 5 - 3sy - 9 + 2y = 4 \frac{1}{s-2}$$

$$s^2 y + 3s - 3sy - 14 + 2y = 4 \frac{1}{s-2}$$

$$s^2 y - 3sy + 2y + 3s - 14 = 4 \frac{1}{s-2}$$

$$s^2 y - 3sy + 2y = -3s + 14 + 4 \frac{1}{s-2}$$

$$y(s^2 - 3s + 2) = -3s + 14 + 4 \frac{1}{s-2}$$

$$y(s^2 - 2s - s + 2) = -3s + 14 + 4 \frac{1}{s-2}$$

$$y\{s(s-2) - 1(s-2)\} = -3s + 14 + 4 \frac{1}{s-2}$$

$$\begin{aligned}
y(s-1)(s-2) &= -3s + 14 + 4 \frac{1}{s-2} \\
y &= -3 \frac{s}{(s-1)(s-2)} + 14 \frac{1}{(s-1)(s-2)} + 4 \frac{1}{(s-2)} \frac{1}{(s-1)(s-2)} \\
y &= \frac{-3s}{(s-1)(s-2)} + \frac{14}{(s-1)(s-2)} + \frac{4}{(s-1)(s-2)^2} \\
y &= \frac{-3s+14}{(s-1)(s-2)} + \frac{4}{(s-1)(s-2)^2} \\
y &= \frac{(-3s+14)(s-2)+4}{(s-1)(s-2)^2} \\
y &= \frac{-3s^2+6s+14s-28+4}{(s-1)(s-2)^2} \\
y &= \frac{-3s^2+6s+14s-24}{(s-1)(s-2)^2}
\end{aligned}$$

Applying partial fraction

Let,

$$y = \frac{-3s^2+6s+14s-24}{(s-1)(s-2)^2} = \frac{A}{(s-1)} + \frac{B}{(s-2)} + \frac{C}{(s-2)^2}$$

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$$\begin{aligned}
y &= \frac{-3s^2+6s+14s-24}{(s-1)(s-2)^2} = \frac{-7}{(s-1)} + \frac{4}{(s-2)} + \frac{4}{(s-2)^2} \\
\therefore L\{Y\} = y &= \frac{-3s^2+6s+14s-24}{(s-1)(s-2)^2} = \frac{-7}{(s-1)} + \frac{4}{(s-2)} + \frac{4}{(s-2)^2} \\
\therefore Y = L^{-1}(y) &= -7 \frac{1}{2} L^{-1}\left[\frac{1}{(s-1)}\right] + 4L^{-1}\left[\frac{1}{(s-2)}\right] + 4L^{-1}\left[\frac{4}{(s-2)^2}\right] \\
\therefore Y = L^{-1}(y) &= -7e^t + 4e^{2t} + 4te^{2t}
\end{aligned}$$

Example 106: Solve the following Initial Value Problem (IVP) by Laplace Transform

$$Y'' + 9Y = \cos 2t \quad Y(0) = 1 \quad Y'\left(\frac{\pi}{2}\right) = -1$$

Solution

$$Y = f(t)$$

Given,

$$Y'' + 9Y = \cos 2t$$

Taking the Laplace transform of both sides of the differential equation and using the given conditions, we have

$$Y'' + 9Y = \cos 2t$$

$$L\{Y''\} + L\{9Y\} = L\{\cos 2t\}$$

We have,

$$L(f''(t)) = s^2 L\{f(t)\} - s f(0) - f'(0)$$

$$\therefore L\{Y''\} + L\{9Y\} = L\{\cos 2t\}$$

$$\therefore L\{Y''\} + 9L\{Y\} = L\{\cos 2t\}$$

$$s^2 L\{f(t)\} - s f(0) - f'(0) + 9y = \frac{s}{s^2 + 4} \quad [\text{let, } L\{Y\} = y]$$

$$s^2 L\{Y\} - s f(0) - f'(0) + 9y = \frac{s}{s^2 + 4}$$

$$s^2 y - s f(0) - f'(0) + 9y = \frac{s}{s^2 + 4} \quad [\text{let, } L\{Y\} = y]$$

$$s^2 y - s \cdot 1 - c + 9y = \frac{s}{s^2 + 4} \quad [\text{let, } f'(0) = c]$$

$$y(s^2 + 9) - s \cdot 1 - c = \frac{s}{s^2 + 4}$$

$$y(s^2 + 9) = s \cdot 1 + c + \frac{s}{s^2 + 4}$$

$$y = \frac{s}{(s^2 + 9)} + \frac{c}{(s^2 + 9)} + \frac{s}{(s^2 + 4)(s^2 + 9)}$$

$$y = \frac{s}{(s^2 + 4)(s^2 + 9)} + \frac{s}{(s^2 + 9)} + \frac{c}{(s^2 + 9)}$$

.....

.....

Applying partial fraction

$$y = \frac{s}{5(s^2 + 4)} - \frac{s}{5(s^2 + 9)} + \frac{s}{(s^2 + 9)} + \frac{c}{(s^2 + 9)}$$

$$y = \frac{s}{5(s^2 + 4)} + \frac{s}{(s^2 + 9)} - \frac{s}{5(s^2 + 9)} + \frac{c}{(s^2 + 9)}$$

$$y = \frac{s}{5(s^2 + 4)} + \frac{5s - s}{5(s^2 + 9)} + \frac{c}{(s^2 + 9)}$$

$$y = \frac{s}{5(s^2 + 4)} + \frac{4s}{5(s^2 + 9)} + \frac{c}{(s^2 + 9)}$$

$$\therefore L\{Y\} = y = \frac{s}{5(s^2 + 4)} + \frac{4s}{5(s^2 + 9)} + \frac{c}{(s^2 + 9)}$$

$$\therefore Y = L^{-1}(y) = L^{-1}\left\{\frac{s}{5(s^2 + 4)}\right\} + L^{-1}\left\{\frac{4s}{5(s^2 + 9)}\right\} + L^{-1}\left\{\frac{c}{(s^2 + 9)}\right\}$$

$$\therefore Y = L^{-1}(y) = L^{-1}\left\{\frac{s}{5(s^2 + 4)}\right\} + L^{-1}\left\{\frac{4s}{5(s^2 + 9)}\right\} + L^{-1}\left\{\frac{c}{(s^2 + 9)}\right\}$$

$$\therefore Y = L^{-1}(y) = \frac{1}{5}L^{-1}\left\{\frac{s}{(s^2+4)}\right\} + \frac{4}{5}L^{-1}\left\{\frac{s}{(s^2+9)}\right\} + \frac{c}{3}L^{-1}\left\{\frac{3}{(s^2+9)}\right\}$$

$$\therefore Y = L^{-1}(y) = \frac{1}{5}L^{-1}\left\{\frac{s}{(s^2+2^2)}\right\} + \frac{4}{5}L^{-1}\left\{\frac{s}{(s^2+3^2)}\right\} + \frac{c}{3}L^{-1}\left\{\frac{3}{(s^2+3^2)}\right\}$$

$$\therefore Y = L^{-1}(y) = \frac{1}{5}\cos 2t + \frac{4}{5}\cos 3t + \frac{c}{3}\sin 3t \dots\dots\dots(i)$$

Given,

$$\therefore Y\left(\frac{\pi}{2}\right) = -1$$

$$\therefore Y = L^{-1}(y) = \frac{1}{5}\cos 2t + \frac{4}{5}\cos 3t + \frac{c}{3}\sin 3t$$

$$\therefore Y\left(\frac{\pi}{2}\right) = \frac{1}{5}\cos 2\left(\frac{\pi}{2}\right) + \frac{4}{5}\cos 3\left(\frac{\pi}{2}\right) + \frac{c}{3}\sin 3\left(\frac{\pi}{2}\right)$$

$$\therefore Y\left(\frac{\pi}{2}\right) = \frac{1}{5}\cos \pi + \frac{4}{5}\cos 3\left(\frac{\pi}{2}\right) + \frac{c}{3}\sin\left(\frac{3\pi}{2}\right)$$

$$\therefore Y\left(\frac{\pi}{2}\right) = \frac{1}{5}(-1) + \frac{4}{5}\cos\left(\frac{3\pi}{2}\right) + \frac{c}{3}\sin\left(\frac{3\pi}{2}\right)$$

$$\therefore Y\left(\frac{\pi}{2}\right) = \frac{1}{5}(-1) + \frac{4}{5} \times 0 + \frac{c}{3}(-1)$$

$$\therefore Y\left(\frac{\pi}{2}\right) = -\frac{1}{5} - \frac{c}{3}$$

$$-1 = \frac{1}{5}(-1) + \frac{c}{3}(-1) \quad [\because Y\left(\frac{\pi}{2}\right) = -1]$$

$$-1 = -\frac{1}{5} - \frac{c}{3}$$

$$\frac{c}{3} = -\frac{1}{15} + 1$$

$$\frac{c}{3} = \frac{-1+15}{15}$$

$$\frac{c}{3} = \frac{14}{15}$$

$$c = \frac{14}{5}$$

Putting the value of c in (i)

$$\therefore Y = L^{-1}(y) = \frac{1}{5}\cos 2t + \frac{4}{5}\cos 3t + \frac{c}{3}\sin 3t \dots\dots\dots(i)$$

$$\therefore Y = L^{-1}(y) = \frac{1}{5}\cos 2t + \frac{4}{5}\cos 3t + \frac{14}{15}\sin 3t$$

Answer

